

Credit Name: CSE 2110 - Procedural Programming 1

Assignment Name: GCF, Hailstone

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I broke down both problems to understand their requirements. For GCF, I needed to find the greatest common factor of two integers. For Hailstone, I needed to repeatedly apply rules to a positive integer until it reached 1.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

How did you decide on the tools, data structures, or algorithms to use?

I planned the steps for each program before coding. For GCF, I used the Euclidean algorithm to find the greatest common factor. For Hailstone, I used a loop and conditional statements to determine whether the current number was even or odd and calculate the next number.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

How did you test your solution along the way to make sure it was working?

I wrote the programs in smaller sections and tested them as I went. I used loops to repeat calculations, conditional statements to handle different cases, and input validation to make sure the programs received appropriate values.

Overcoming Challenges

What part of the problem was the most difficult for you?

How did you handle moments when you felt stuck or unsure of what to do next?

One of the most difficult parts was making sure the loops stopped at the correct time. I tested different starting values and pairs of integers to check that GCF returned the correct result and Hailstone reached 1 while counting the steps accurately.

Learning

Was there anything you learned that you think will help you with future challenges?

I learned more about using algorithms, loops, conditional statements, and validation in Java. I also gained a better understanding of how mathematical problems can be broken down into smaller steps and solved through programming.